

Distributional Reinforcement Learning for Multi-Dimensional Reward Functions

Summary

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1 Problem Setup

Traditional reinforcement learning assumes scalar rewards, but many real-world problems involve multiple, potentially correlated reward signals. For example, autonomous driving must balance safety, speed, comfort, and efficiency simultaneously. This paper introduces distributional RL for multi-dimensional rewards, where the agent learns the joint probability distribution of returns across multiple dimensions rather than just their expectations.

2 Thesis

The central thesis is that learning the joint distribution of multi-dimensional rewards provides crucial information about correlations and trade-offs between different objectives that is lost when treating rewards independently. The authors argue that modeling these correlations is essential for optimal decision-making in complex, multi-objective environments.

3 Key Concepts

Multi-Dimensional Return Distribution: The joint probability distribution over all possible cumulative future rewards across multiple dimensions from a given state.

Joint Distributional Bellman Operator: A generalization of the distributional Bellman operator that operates on joint probability distributions, capturing correlations between different reward dimensions.

MD3QN (Multi-Dimensional Distributional DQN): The proposed algorithm that learns joint return distributions using a neural network architecture designed to capture inter-dimensional correlations.

Maximum Mean Discrepancy (MMD): A metric used to measure the distance between the predicted and true joint distributions, providing a training objective for distributional learning.

Correlated Constraints: Multiple constraints that are statistically dependent, requiring joint distribution modeling to properly assess the probability of satisfying all constraints simultaneously.

4 Methods

The authors develop several key methodological contributions:

1. **Joint Distribution Modeling:** They extend distributional RL to handle multi-dimensional rewards by learning joint probability distributions rather than marginal distributions.

2. **MD3QN Algorithm:** They introduce a neural network architecture that outputs samples from the joint return distribution, using specialized network designs to capture correlations.
3. **MMD-Based Training:** They use Maximum Mean Discrepancy as the training objective to match the predicted joint distribution with the true distribution.
4. **Correlation-Aware Decision Making:** They demonstrate how the learned joint distributions can be used to make better decisions when multiple constraints must be satisfied simultaneously.

The key innovation is the ability to model and exploit correlations between different reward dimensions, which is impossible with approaches that only learn marginal distributions.

5 Results

The authors demonstrate the effectiveness of their approach on several domains:

- **Maze Environments:** In custom grid worlds with multiple reward types, MD3QN accurately captures the joint distribution and correlations between different reward sources.
- **Atari Games:** The algorithm successfully models joint distributions in complex environments like AirRaid, Asteroids, and Gopher, capturing both reward correlations and appropriate scales.
- **Multi-Constraint Optimization:** In environments with multiple correlated constraints, MD3QN significantly outperforms baselines that only model marginal distributions.
- **Distribution Accuracy:** The MMD metric shows that MD3QN achieves much lower error in approximating the true joint distribution compared to alternative approaches.
- **Decision Quality:** When optimizing for the probability of satisfying multiple constraints, joint distribution modeling leads to substantially better policies than approaches that ignore correlations.

The results show that ignoring correlations between reward dimensions can lead to suboptimal or even dangerous policies in real-world applications.

6 Significance

This work extends distributional reinforcement learning to the multi-dimensional setting with several important implications:

- **Real-World Applications:** Provides a framework for handling complex multi-objective problems like autonomous driving, robotics, and resource management where multiple correlated objectives must be balanced.
- **Risk Assessment:** Enables more sophisticated risk assessment by modeling how different failure modes or constraints are correlated, rather than assuming independence.
- **Theoretical Foundation:** Establishes the mathematical foundations for multi-dimensional distributional RL, extending the contraction properties of the Bellman operator to joint distributions.
- **Practical Algorithm:** Provides a practical algorithm (MD3QN) that can be applied to real-world problems with multiple reward signals.

- **Safety-Critical Systems:** Particularly important for safety-critical applications where understanding the joint probability of multiple constraints being violated is crucial for safe operation.

The paper demonstrates that in many real-world scenarios, the correlations between different objectives are not just a minor detail but a fundamental aspect of the problem that must be modeled for optimal and safe decision-making.